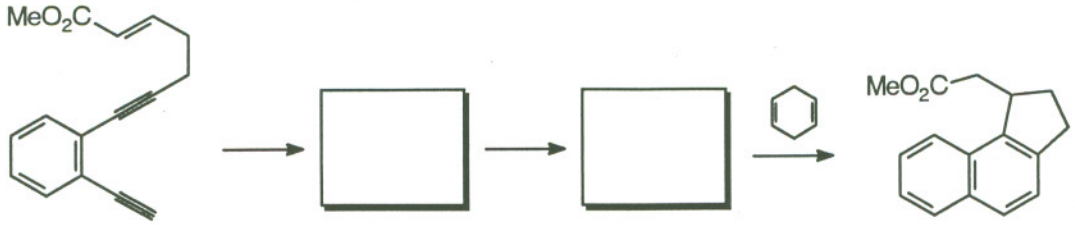
Organic Chemistry Cumulative Examination
15 January 1994

1. Fill in the missing intermediate structures (the empty boxes in the middle) in the following reaction sequence. Also show how you get from one structure to the next using the usual curved-arrow–electron pushing notation.

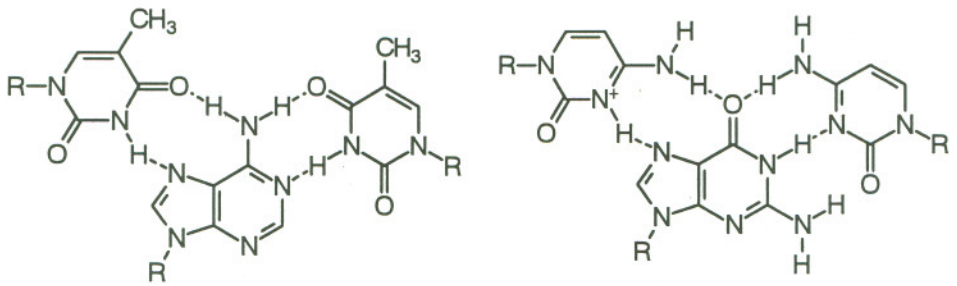


(*J. Am. Chem. Soc.* **1993**, *115*, 11744)

2. Same as no.1 above, for the followint reaction sequence.

(*J. Org. Chem.* **1993**, *58*, 7320)

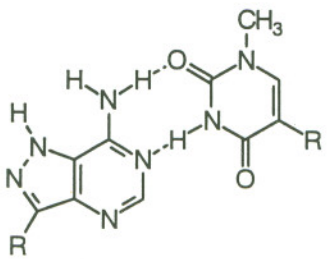
3. The ability of nucleotides to form base triplets (e.g., T-A-T and C^+-G-C, shown below) have been utilized in the design of oligonucleutides for sequence-specific binding to DNA, as recently discussed in a departmental seminar by Dr. Kenneth Turnbull.



T-A-T $\qquad\qquad$ C^+-G-C

As you may recall, Dr. Turnbull's efforts are directed towards the design and synthesis of synthetic nucleotides which will serve to bind to base pairs other than A-T and G-C, namely C-G and T-A.

In a different context, Dr. Steven Benner and his colleagues at the ETH have been designing synthetic nucleotides for incorporation into double-helical DNA. One example is the formycin A–5-methylpseudouridine (F-Ψ) base pair (*Biochemistry* **1991**, *30*, 10351).



F-Ψ

Using the same strategy as exemplified in T-A-T and C^+-G-C (or in Dr. Turnbull's work), design a nucleotide base that will serve to bind the F-Ψ base pair. In your design, take care to account for any possible tautomerism which may prevent the expected binding interaction.